

Curriculum Vitae

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Name: Natalia Chepiga
Nationality: Ukrainian, Dutch
Place of birth: Kharkiv, Ukraine
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Address: Rudolf Peierls Centre for Theoretical Physics,
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Languages: English, Ukrainian, Russian, (all fluent), French(B1), Dutch(B1),
German(A2)



Expertise:

Computational physics, condensed matter physics, quantum many-body physics and strongly-correlated systems, tensor networks, quantum phase transitions, conformal field theory, quantum simulators, quantum magnetism, chiral melting, constrained systems (non-abelian anyons, quantum dimers and quantum loops, supersymmetric fermionic models), low-dimensional quantum systems, Rydberg atoms, topological phases, systems with multi-component Hilbert space, comb tensor networks, disorder, infinite randomness

Education:

- 04/13 – 04/17 Docteur ès sciences, Institute of Physics, École Polytechnique Fédérale de Lausanne,
Supervisor: prof. Frédéric Mila
Thesis Title: ***Dimerization and exotic criticality in spin-S chains***
Private defense: 21/02/2017; Public defense: 23/03/2017
Distinction from Doctoral School of Physics, EPFL
- 08/11 – 02/13 Master in Physics, École Polytechnique Fédérale de Lausanne,
Supervisor: prof. Frédéric Mila
Thesis Title: ***Topological phase transitions in spin ladders***
- 09/07 – 07/11 BSc in Applied Physics with First Class Honors, V.N.Karazin Kharkiv National
University, Department of Theoretical Nuclear Physics
Supervisor: Sergey I. Shevchenko; Thesis Title: *Description of the electrons-holes
superfluidity in terms of the order parameter*
- 09/00 – 06/07 High School Certificate with First Class Honors

Employment:

- 06/26-now **Research Fellow** at Somerville College, Oxford, UK
04/26-now **Royal Society University Research Fellow** at the University of Oxford, UK
- 05/24-now **Associate Editor** of the Physical Review Research of American Physical Society
- 02/26-03/26 **ERC Research Fellow** at the University of Oxford, UK
01/21-01/26 **Assistant professor**, Kavli Institute of Nanoscience, Faculty of Applied Sciences,

Delft University of Technology, Netherlands

- 01/19-12/20 **Postdoc** in the group of **prof. P. Corboz** at the University of Amsterdam, Netherlands. The work has been supported by the Swiss National Science Foundation (grant number P400P2_183847) and by prof. Corboz's funds.
- 05/17-12/18 **Postdoc** in the group of **prof. S.R. White** at the University of California, Irvine, USA. The work has been supported by the Swiss National Science Foundation (grant number P2ELP2_172271) and by prof. White's funds.
- 04/13 – 04/17 **Doctoral assistant** at the Chair of condensed matter theory, Institute of Physics, École Polytechnique Fédérale de Lausanne, Supervisor: prof. Frédéric Mila

Selected Awards:

- 01/23-12/24 **Visiting professor**, Université Paul Sabatier, Toulouse, France
- 11/21 **Minerva prize** by Dutch Physics Council and Netherlands' Physical Society (<https://dutchphysicscouncil.nl/613-4>)
- 12/17 **Distinction from the Doctoral School of Physics**, EPFL for the thesis *Dimerization and exotic criticality in spin-S chains*
- 10/11 – 02/13 **Excellence scholarship** provided by École Polytechnique Fédérale de Lausanne
- 09/07 – 06/11 Government scholarships for university students with outstanding results
- 09/06 – 08/08 **2xPresident of Ukraine Scholarships**
- 09/03 – 06/11 Several diploma including 1st and 2nd prizes in Olympiads in Physics; 1st prize in Ukrainian Competition of Research projects

Grants and funding (personal):

- 04/26-03/34 **Royal Society University Research Fellowship GBP 1.94M** “Advancing the theory of quantum phase transitions in the era of quantum computing”
- 02/26-01/31 **ERC starting Grant EUR 1.92M**, from the European Research Council for the project “TRANGINEER”.
- 01/24-12/25 USD 123k from **Julian Schwinger Foundation (USA)** for the project “Challenging the theory of Mott transitions”
- 02/19-09/20 PostdocMobility by the **Swiss National Science Foundation**, University of Amsterdam, The Netherlands. Project title: Further development of infinite Projected Entangled Pair States (iPEPS): network of clusters and hard constraints
- 04/17-09/18 EarlyPostdocMobility by the **Swiss National Science Foundation**, University of California, Irvine, USA. Project title: Efficient Density Matrix Renormalization Group (DMRG) algorithm for two-dimensional systems and its applications.
- 08/25 1M CPU hours by SURFSARA national supercomputing cluster Snellius (EINF-15112)
- 01/24 1M CPU hours by SURFSARA national supercomputing cluster Snellius (EINF-8242)
- 02/23 GBP 9.5k from IQTN/EP SRC for the workshop “Tensor networks for constrained systems”;
- 08/22 1M CPU hours by SURFSARA national supercomputing cluster Snellius (EINF 3879);
- 02/22 100k CPU hours by SURFSARA national supercomputing cluster Lisa (EINF 2722);

02/22 Aspasia EUR 120k; Awarded by Dutch Research council **NWO**; not accepted by TUDelft.
02/21 500k CPU hours by SURFSARA national supercomputing cluster Cartesius (EINF 1137)
02/21 100k CPU hours by SURFSARA national supercomputing cluster Lisa (EINF 1137)

Collective grants and networks:

10/26-09/31 Topic leader in the **program grant by EPSRC** “Fundamental Science in the Age of Quantum Computing”, led by A.G.Green; GBP 7.3M

10/25-09/28 Collaborator on the Spanish National Grant “TENSOR networks, Temporal entanglement, and the Real-time dynamics of out-of-Equilibrium quantum System” led by L.Tagliacozzo; EUR 192k.

04/2025- “Entanglement scaling and quantum phases in a superconducting quantum dot chain”, together with M.Wimmer (TUDelft); EUR 317k from SUMMIT-Quantum Limits (NWO)

08/2023 – now **The Kavli innovation award**: a consortium of 13 PI at TUDelft; USD 5M
<https://www.tudelft.nl/en/2023/tnw/5-million-in-quest-for-missing-link-in-quantum-communication>

10/22-now “Materials for the quantum age”, a consortium of 43 PIs and 34 PhD and postdocs, supported by Dutch research council (NWO), <https://qumat.org/people/>

02/22-now Partner of the **International Quantum Tensor Networks**, (seeding funds from EPSRC)
<https://iqtn.phys.strath.ac.uk/>

01/2021-now Member of the **European Tensor Network** (quantumtensor.pks.mpg.de)

2013-2017 Member of **MaNEP** network and Swiss National Science Foundation

Publications:

44. J.Colbois, **N.Chepiga**, S.Adam, G.Lemarié, N.Laflorencie,
Extreme statistics as a probe of the superfluid to Bose-glass Berezinskii-Kosterlitz-Thouless transition;
New J. Phys 28, 083510 (2026)
43. Julien Fitouchi, **Natalia Chepiga**,
Commensurate-incommensurate Mott transition without magnetic field: emergence of nematic Luttinger liquid in XXZ chain
Phys. Rev. Lett. 137, 036503 (2026)
42. Y.Liu, **N.Chepiga**, Y.Fukusumi, M.Oshikawa,
Boundary critical phenomena in the quantum Ashkin-Teller model, arXiv:2601.16951,
SciPost Physics (2026)
41. Bowyer La Riviere, **Natalia Chepiga**,
Extended Ashkin-Teller transition in two coupled frustrated Haldane chains,
Phys. Rev. B 113, 205107 (2026)
40. Jose Soto Garcia, **Natalia Chepiga**,
The quantum Kibble-Zurek mechanism: the role of boundary conditions, endpoints and kink types
Phys. Rev. B 113, 085430 (2026)

39. Aman Sharma, Mithilesh Nayak, **Natalia Chepiga**, Henrik M. Rønnow, Frédéric Mila,
Confinement, deconfinement, and bound states in the spin-1 and spin-3/2 generalizations of the Majumdar-Ghosh chain
Phys. Rev. B 113, 054438 (2026)
38. Jose Soto Garcia, **Natalia Chepiga**,
Infinite randomness criticality and localization of the floating phase in arrays of Rydberg atoms trapped with non-perfect tweezers
Phys. Rev. Lett. 136, 056502 (2026)
37. Aman Sharma, Mithilesh Nayak, **Natalia Chepiga**, and Frédéric Mila
Excitations and dynamical structure factor of J1-J2 spin-3/2 and spin-5/2 Heisenberg spin chains
Phys. Rev. B 112, 104401 (2025)
36. Niels T. Pronk, Bowe La Riviere, **Natalia Chepiga**
Deconfined quantum criticality in a frustrated Haldane chain with single-ion anisotropy
Phys. Rev. B **Letter** 111, L220412 (2025), **Editor's Suggestion**
35. Jose Soto Garcia, **Natalia Chepiga**,
Numerical investigation of quantum phases and phase transitions in a two-leg ladder of Rydberg atoms
Phys. Rev. Research 7 (1), 013215 (2025)
34. Bowe La Riviere, **Natalia Chepiga**,
Z4 transitions in quantum loop models on a zig-zag ladder
SciPost Phys. 17, 144 (2024)
33. **Natalia Chepiga**,
Probing universal critical scaling with scan-DMRG
Phys. Rev. B 110, 144401 (2024)
32. Jose Soto Garcia, **Natalia Chepiga**,
Resolving chiral transition in Rydberg arrays with quantum Kibble-Zurek mechanism and finite-time scaling
Phys. Rev. B 110, 125113 (2024)
31. **Natalia Chepiga**,
Realization of Wess-Zumino-Witten transitions with levels $k=6$ and $k=4$ in a frustrated spin-3 chain;
Phys. Rev. B 109, 214403 (2024)
30. **Natalia Chepiga**,
Tunable quantum criticality in multi-component Rydberg arrays;
Phys. Rev. Lett. 132, 076505 (2024)
29. **Natalia Chepiga**, Nicolas Laflorencie,
Resilient infinite randomness criticality for a disordered chain of interacting Majorana fermions;
Phys. Rev. Lett. 132, 056502 (2024)
28. Bernhard Lüscher, Frederic Mila, **Natalia Chepiga**,
Critical properties of the quantum Ashkin-Teller chain with chiral perturbations;
Phys. Rev. B 108, 184425 (2023)
27. Zakaria Jouini, **Natalia Chepiga**, Loic Herviou, Frederic Mila,

Emergent $U(1)$ symmetry in non-particle-conserving 1D models;
Phys. Rev. B 108, 205145 (2023)

26. **Natalia Chepiga**,
Critical properties of the Majorana chain with competing interactions;
Phys. Rev. B 108, 054509 (2023)
25. **Natalia Chepiga**, Nicolas Laflorencie,
Topological and quantum critical properties of the interacting Majorana chain;
SciPost Phys. 14, 152 (2023)
24. **Natalia Chepiga**, Frédéric Mila,
Eight-vertex criticality in the interactive Kitaev chain;
Phys. Rev. B 107, L081106 (2023)
23. **Natalia Chepiga**,
From Kosterlitz-Thouless to Pokrovsky-Talapov transitions in spinless fermions and spin chains with next-nearest-neighbor interactions;
Phys. Rev. Research 4, 043225 (2022)
22. Ivo A. Maceira, **Natalia Chepiga**, Frédéric Mila,
Conformal and chiral phase transitions in Rydberg chains;
Phys. Rev. Research 4, 043102 (2022)
21. **Natalia Chepiga**,
Critical properties of quantum three- and four-state Potts models with boundaries polarized along the transverse field
SciPost Phys. Core 5, 031 (2022)
20. **Natalia Chepiga**, Ian Affleck, Frédéric Mila,
From $SU(2)_5$ to $SU(2)_3$ Wess-Zumino-Witten transitions in a frustrated spin-5/2 chain
Phys. Rev. B 105, 174402 (2022); **Editors' Suggestion**
19. **Natalia Chepiga**, Jiří Minář, Kareljan Schoutens,
Supersymmetry and multicriticality in a ladder of constrained fermions
SciPost Phys. 11, 059 (2021)
18. **Natalia Chepiga** and Frédéric Mila,
Lifshitz point at commensurate melting of 1D Rydberg atoms
Phys. Rev. Research, 3, 023049 (2021)
17. **Natalia Chepiga** and Frédéric Mila,
Kibble-Zurek exponent and chiral transition of the period-4 phase of Rydberg chains
Nature Communications, 12, 414 (2021)
16. Mario Motta, Claudio Genovese, Fengjie Ma, Zhi-Hao Cui, Randy Sawaya, Garnet Kin-Lic Chan, **Natalia Chepiga**, Phillip Helms, Carlos Jimenez-Hoyos, Andrew J. Millis, Ushnish Ray, Enrico Ronca, Hao Shi, Sandro Sorella, Edwin M. Stoudenmire, Steven R. White, Shiwei Zhang (Simons collaboration on the many-electron problem)
Ground-state properties of the Hydrogen chain: insulator-to-metal transition, dimerization, and magnetic phases
Phys. Rev. X 10, 031058 (2020)

15. **Natalia Chepiga**, Steven R. White,
Critical properties of a comb lattice
SciPost Phys. 9, 013 (2020)
14. **Natalia Chepiga**, Ian Affleck, and Frédéric Mila,
Floating, critical, and dimerized phases in a frustrated spin-3/2 chain
Phys. Rev. B 101, 174407 (2020)
13. Laurens Vanderstraeten, Elisabeth Wybo, **Natalia Chepiga**, Frank Verstraete, and Frédéric Mila,
Spinon confinement and deconfinement in a spin-1 chain
Phys. Rev. B 101, 115138 (2020);
12. **Natalia Chepiga** and Frédéric Mila,
Dimerization and effective decoupling in two spin-1 generalizations of the spin-1/2 Majumdar-Ghosh chain
Phys. Rev. B 100, 104426 (2019);
11. **Natalia Chepiga** and Steven R. White,
Comb tensor networks
Phys. Rev. B 99, 235426 (2019)
10. **Natalia Chepiga** and Frédéric Mila,
DMRG investigation of constrained models: from quantum dimer and quantum loop ladders to hard-boson and Fibonacci anyon chains
SciPost Phys. 6, 033 (2019);
9. **Natalia Chepiga** and Frédéric Mila,
Floating phase versus chiral transition in a 1D hard-boson model
Phys. Rev. Lett. 122, 017205 (2019)
8. **Natalia Chepiga** and Frédéric Mila,
Rigorous decoupling between edge states in frustrated spin chains and ladders
Phys. Rev. B 97, 174434 (2018)
7. **Natalia Chepiga** and Frédéric Mila,
Exact zero modes in frustrated Haldane chain
Phys. Rev. B 96, 060409 (2017), **Rapid Communication**
6. **Natalia Chepiga** and Frédéric Mila,
Excitation spectrum and Density Matrix Renormalization Group iterations
Phys. Rev. B 96, 054425 (2017)
5. L.Wang, **N.Chepiga**, D.-K.Ki, L.Li, F.Li, W.Zhu, Y.Kato, O.S.Ovchinnikova, F.Mila, I.Martin, D.Mandrus, A.F.Morpurgo,
Controlling the topological sectors of magnetic solitons in exfoliated $\text{Cr}_{1/3}\text{NbS}_2$ crystals
Phys. Rev. Lett. 118, 257203 (2017), Editor's Suggestion
4. **Natalia Chepiga**, Ian Affleck, and Frédéric Mila,
Spontaneous dimerization, critical lines, and short-range correlations in a frustrated spin-1 chain
Phys. Rev. B 94, 205112 (2016)

3. **Natalia Chepiga**, Ian Affleck, and Frédéric Mila,
Comment on "Frustration and Multicriticality in the Antiferromagnetic Spin-1 Chain"
Phys. Rev. B 94, 136401 (2016)
2. **Natalia Chepiga**, Ian Affleck, and Frédéric Mila,
Dimerization transitions in spin-1 chains
Phys. Rev. B 93, 241108 (2016), **Rapid Communication**
1. **Natalia Chepiga**, Frédéric Michaud, and Frédéric Mila,
Berry phase investigation of spin-S ladders
Phys. Rev. B 88, 184418 (2013)

Pre-prints:

1. Manu Canals, **Natalia Chepiga**, Luca Tagliacozzo, A tensor network formulation of Lattice Gauge Theories based only on symmetric tensors; arXiv:2412.16961
2. Bowy La Riviere, Rik Mulder, **Natalia Chepiga**, Exact zero modes in the interacting Majorana X- and Y-junctions, arXiv: 2506.10898
3. Jose Soto Garcia, **Natalia Chepiga**, Long-lived revivals and real-space fragmentation in chains of multispecies Rydberg; arXiv:2604.13257
4. Bowy La Riviere, **Natalia Chepiga**, Non-magnetic floating phases in frustrated Haldane chains with a single-ion anisotropy, arXiv:2604.25493

Invited conference talks (36+1 planned):

- 12/26 (upcoming) TNSAA, RIKEN, Japan,
- 09/26 Quantum and Topological in Real Materials, France, *Quantum criticality... trapped with non-perfect tweezers*
- 07/26 Condensed Matter Physics in the City 2026, **UK**, *Non-magnetic floating phases in Haldane chains*
- 07/26 Exotic Quantum Phases in Equilibrium and Driven Matter, Zanzibar, **Tanzania**,
Non-magnetic floating phases in Haldane chains
- 02/26 Quantum Workshop Coventry 2026, UK, *Quantum criticality... trapped with non-perfect tweezers*
- 11/25 **Keynote** at Øredev Developers conference, Malmö, **Sweden**, *Networking for quantum: how simulations help us to design the future*
- 10/25 COOLMAG 2025, Košice, **Slovakia**, *"C-IC transition without magnetic field"*
- 10/25 Delta-ITP meeting, Amsterdam, Netherlands, *"Probing quantum criticalities with Kibble-Zurek mechanism"*
- 09/25 ICTP-SAIFR workshop, São Paulo, **Brazil**, *"Probing quantum criticalities with Kibble-Zurek mechanism"*
- 05/25 Entanglement in Many-body Quantum Matter: Dynamics, Dissipation, Equilibration, **ESI, Vienna**, Austria, *Tensor network perspectives of quantum Kibble-Zurek mechanism*
- 05/25 Quantum and Networking (DFG Research Unit FOR 5522), Göttingen, Germany, *Scan-DMRG*
- 02/25 Aquemics : Advanced quantum materials for magnetic cooling and quantum information science, Cergy, France, *Dimerization transition in spin-S chains*
- 05/24 Theories, Experiments and Numerics on Gapless Quantum Many-body Systems, **KITP, Santa Barbara**, USA, *Tunable quantum criticality in multi-component Rydberg arrays*
- 05/24 "Bridging the Gap between Classical & Quantum Simulation", **Lorentz Center**, The Netherlands;

- 04/24 *Tunable quantum criticality: Challenging quantum computers with classical simulations.*
TUM-IAS workshop, Garching, Germany, *Resilient infinite randomness for interacting Majorana fermions*
- 01/24 **Plenary meeting of the International Quantum Tensor Network**, Glasgow, UK,
Tunable quantum criticality in multi-component Rydberg arrays
- 11/23 "Chaos and information dynamics in quantum many-body systems", **Ettore Majorana Center**,
Erice, Sicily, *Resilient infinite randomness criticality for interacting Majorana fermions*
- 11/23 "Quantum information: theory and applications", Paris, France, *Tunable quantum criticality
in multi-component Rydberg arrays*
- 09/23 "Quantum many-body methods in cond-mat systems", **RWTH Aachen**, Germany, *The power
of Friedel oscillations. Critical properties of interacting Majorana chains*
- 09/23 **Korrelationstage 2023**, Dresden, Germany, *Resilient infinite randomness criticality for a
disordered chain of interacting Majorana fermions*
- 08/23 **NG SCES 2023**, Lido di Fermo, Italy, *9 ½ phases of interacting Majorana chains*
- 08/23 Entanglement in strongly correlated systems, **Benasque**, Spain, *Critical properties of
interacting Majorana fermions*
- 08/23 The Grete Hermann Network Workshop, Wuerzburg, Germany, *When Kosterlitz and
Thouless meet Pokrovsky and Talapov*
- 07/23 JSF Workshop on the fermion sign problem, **Peyresq**, France, *An odd sequence of WZW
criticalities in a frustrated spin-5/2 chain*
- 06/23 "Quantum Materials: Experimental Enigmas and Theoretical Challenges", **Aspen**, USA,
When Kosterlitz and Thouless meet Pokrovsky and Talapov – a computational enigma
- 06/23 "Exotic Phases, Gauge Field Theories and Dynamics in Systems with Constraints", **Aspen**,
USA, *Introduction to Constrained tensor networks*
- 11/22 "Entanglement Scaling and Criticality with Tensor Networks", **Bernoulli Center**, Switzerland,
Critical properties of an interacting Majorana chain. The power of Friedel oscillations
- 10/22 "Symmetry and Duality in Quantum Many-Body Systems," Ghent, Belgium, *Dual boundary
conditions in minimal models*
- 09/22 "Computational aspects of Tensor Networks", **Erwin Schrodinger International Institute**, Austria,
Eight vertex criticality in interacting Kitaev chains
- 01/22 **Physics@Veldhoven**, *Lifshitz point or Why the transition becomes chiral?*
- 10/21 **CECAM flagship** workshop: Computational materials discovery of unconventional magnets,
Lausanne, Switzerland, *Floating phases in quantum spin chains*
- 02/21 Entanglement in Strongly Correlated Systems, Benasque, Spain, *Supersymmetric point in a
ladder of constrained fermions*
- 12/20 European Tensor Network online series, *Chiral transitions in chains of Rydberg atoms*
- 11/19 Delta-ITP triangle meetings: Quantum and Topological Matter, University of Utrecht,
The Netherlands, *Comb tensor networks*
- 03/19 **DPG Frühjahrstagung 2019**, Regensburg, Germany, *DMRG investigation of constrained
models: from quantum dimer and quantum loop ladders to hard-boson and Fibonacci anyons*
- 02/19 Constrained Many-body Dynamics, **MPI PKS, Dresden**, Germany, *DMRG investigation of
constrained models: from quantum dimer and quantum loop ladders to hard-boson and
Fibonacci anyon Quantum many-body chains*
- 06/18 TOPMAT, Paris-Saclay, France, *DMRG investigation of quantum dimer ladders*

Invited lectures at PhD schools (12+ planned):

- 09/26 MPI-PKS school "From quantum matter to quantum computers" Dresden
- 05/26 Colloquium+Lecture+Editorial Insight at ALPS III, Les Diablerets, Switzerland
- 11/25 European Tensor Network School, Goettingen, **Germany** (whole day)
- 08/25 Graduate Lectures "Introduction to Tensor Networks", **Weizmann institute**, Israel

- 03/25 Distinguished lecturer for Quantum Digital Twins Masterclass, **NPL, UK**, "*Constrained Tensor Networks and what one can do with them*"
- 09/24 **Les Houches** School "Frontiers of Condensed Matter"
- 06/24 8th **Les Houches** School in Computational Physics: Variational Approaches for quantum matter in and out of equilibrium, "*Constrained Tensor Networks and what one can do with them*"
- 09/23 Topological Quantum Matter School, Leipzig, **Germany**, *Chiral transitions in Rydberg atoms*
- 09/23 European Tensor Network school, Abingdon, **UK**, *Introduction to MPS (whole day)*
- 05/23 DRSTP condensed matter theory school, Callantsoog, **The Netherlands**; *Quantum phase transitions (5 lectures, 1.5h each)*
- 04/23 lecture at the JuniorClub at the University of Paul Sabatier Toulouse, **France**, *bCFT with DMRG*
- 07/19 Computational Approaches to Quantum Many-body Problems, ISSP, Kashiwa, **Japan**, *Practical introduction to MPS + Comb tensor networks + DMRG for constrained models (2 lectures)*

Invited seminars and colloquia (40 + 3 planned):

- 11/26(upcoming) **Bristol**
- 10/26(upcoming) Atoms and lasers seminar (**Oxford**); *Long-lived revivals in multispecies Rydberg arrays*
- 9/26 (upcoming) **UCL**; *When Pokrovsky and Talapov meet Kosterlitz and Thouless*
- 09/26 **EPFL**, Switzerland; *Quantum criticality trapped with non-perfect tweezers*
- 07/26 Atoms and lasers seminar (**Oxford**); *Quantum criticality trapped with non-perfect tweezers*
- 03/26 **Cambridge**, UK; *The saga of chiral transitions in Rydberg arrays*
- 02/26 Condensed Matter Physics, Oxford; *When Pokrovsky and Talapov meet Kosterlitz and Thouless*
- 02/26 Condensed Matter Theory Forum **Oxford**; *The saga of chiral transitions in Rydberg arrays*
- 11/25 **Sorbonne** University (France); *Scan-DMRG*
- 11/25 Institute Pacal, Paris-Saclay (France); *Quantum criticality with non-perfect tweezers*
- 04/25 University of **Tuebingen**, Germany;
colloquium *Tunable phase transitions in Rydberg-based quantum simulators*
- 02/25 iQuISE Seminar at **MIT, USA**; *Tunable phase transitions in Rydberg-based quantum simulators*
- 02/25 University of Strathclyde, **Glasgow**, UK;
colloquium *Tunable phase transitions in Rydberg-based quantum simulators*
- 11/24 **EPFL**, Switzerland; *Probing universal critical scaling with scan-DMRG*
- 10/24 University of **Zurich**, Switzerland; *Probing universal critical scaling with scan-DMRG*
- 10/24 University of **Innsbruck**, Austria; *Tunable quantum criticality in Rydberg atoms*
- 09/24 **CNRS** LPT Toulouse, France, *Probing universal critical scaling with scan-DMRG*
- 03/24 Kharkiv National University, Ukraine; *Tunable quantum criticality in Rydberg atoms*
- 03/24 **Flatiron Institute**, USA; *Tunable quantum criticality in Rydberg atoms: challenging quantum simulators with classical computers*
- 03/24 **CEA-Saclay**, France; *Tunable quantum criticality in Rydberg atoms*
- 01/24 **TUWien**, Austria; *Tunable chiral transitions in Rydberg atoms*
- 01/24 University of **Goettingen**, Germany; host: S.Manmana;
(Tunable) chiral transitions in Rydberg atoms
- 09/23 Seminar at **CNRS** LPT Toulouse, France, *(Tunable) chiral transitions in Rydberg atoms*
- 05/23 University of **Geneva**, Switzerland, *Critical properties of the interacting Majorana chains*
- 04/23 **CNRS** LPT Toulouse, France, *Critical properties of the interacting Majorana chains*
- 01/23 Vision Seminar at TUDelft; *When Kosterlitz and Thouless meet Pokrovsky and Talapov*
- 11/22 **CNRS** LPT Toulouse; host: Nicolas Laflorencie; *When Kosterlitz and Thouless meet Pokrovsky and Talapov*
- 11/22 Seminar at **Nijmegen**, The Netherlands; host: A.Bagrov; *When Kosterlitz and Thouless*

- meet Pokrovsky and Talapov
- 06/22 Physical Sciences Seminar at **ISTA**, Austria; host: Maksym Serbyn; *Supersymmetry and multicriticality in a ladder of constrained fermions*
- 06/22 **TUWien**, Austria; host: Julian Leonard; *Chiral transitions in chains of Rydberg atoms*
- 12/21 Utrecht Condensed Matter Theory Seminar, NL, *Chiral transitions in chains of Rydberg atoms*
- 10/21 **Brookhaven National Laboratory**, USA, *Probing conformal towers of states with Density Matrix Renormalization Group algorithms*
- 05/21 **ETH Zurich**, Switzerland; *Chiral transitions in chains of Rydberg atoms*
- 02/21 **Harvard** Condensed Matter Theory Seminar, USA; *Chiral transitions in chains of Rydberg atoms*
- 12/20 University of **Amsterdam**, The Netherlands; host: P.R. Corboz, *Constrained tensor networks: a new approach to quantum criticality*
- 02/20 TU Delft, The Netherlands; *Tensor network investigation of constrained models: from quantum dimer and quantum loop ladders to chains of Rydberg bosons*
- 06/19 **University of Nottingham**, UK; host: Juan P. Garrahan, *Constrained DMRG as a byway to investigate critical properties of frustrated magnets*
- 02/19 University of Amsterdam, The Netherlands; host: P.R. Corboz, *Floating phase versus chiral transition in constrained models*
- 10/18 University of California, **Irvine**, USA; host: Steven R. White, *Floating phase versus chiral transition in constrained models: from hard-boson chain to quantum dimer and quantum loop ladders*
- 03/18 HISKP, Universität **Bonn**, Germany; host: Corinna Kollath, *DMRG investigation of quantum dimer ladders*
- 02/18 **Max-Planck-Harvard** Institute for Quantum Optics, Garching, Germany; host: Ignacio Cirac, *Frustrated spin chains: exotic criticality, exact zero modes and quantum dimer model.*
- 11/17 **Perimeter Institute**, Waterloo, Canada; host: Guifre Vidal, *Spontaneous dimerization, critical lines and exact zero modes in frustrated spin-1 chain.*
- 10/17 University of **British Columbia**, host: Ian Affleck, *Exact zero modes in frustrated spin chains*

Contributed talks (19):

- 07/22 International conference on strongly correlated electron systems (**SCES**) 2022, Amsterdam, The Netherlands, *From $SU(2)_5$ to $SU(2)_3$ Wess-Zumino-Witten transitions in a frustrated spin-5/2 chain*
- 06/22 Highly Frustrated Magnetism (**HFM**) 2022, Paris, France, *From $SU(2)_5$ to $SU(2)_3$ Wess-Zumino-Witten transitions in a frustrated spin-5/2 chain*
- 01/21 Waiting for Highly Frustrated Magnetism 2021, Dresden, Germany, *Floating, critical and dimerized phases in a frustrated spin-3/2 chain*
- 12/20 Exploring quantum many-body physics with ultra-cold atoms and molecules, Bad Honnef, Germany, *Kibble-Zurek exponent and chiral transition of the period-4 phase of Rydberg chains*
- 02/20 Entanglement in Strongly Correlated Systems, Benasque, Spain, *Ashkin-Teller transition of Rydberg atoms with two-site blockade*
- 01/20 Physics@Veldhoven 2020, Veldhoven, The Netherlands, *Simulating constrained models with tensor networks*
- 09/19 Korrelationstage 2019, Dresden, Germany, *Comb tensor networks*
- 10/18 Topological phases in condensed matter and cold atom systems, Cargese, Corsica, France *Constrained DMRG as a byway to investigate critical properties of frustrated magnets*
- 06/18 Trends in quantum magnetism, Bad Honnef, Germany, *DMRG investigation of quantum dimer ladders*
- 02/18 Entanglement in Strongly Correlated Systems, Benasque, Spain,

- DMRG investigation of quantum dimer ladders*
- 11/17 Novel Quantum States in Condensed Matter 2017, Kyoto, Japan, *Spontaneous dimerization, critical lines, and exact zero modes in a frustrated spin-1 chain*
 - 06/17 Many Electron Collaboration Summer School, Stony Brook, USA, *Excitation spectrum and Density Matrix Renormalization Group iterations*
 - 02/17 Entanglement in strongly correlated systems, Benasque, Spain, *Dimerization and exotic criticality in spin-S chains*
 - 09/16 Recent Progress in Low-Dimensional Quantum Magnetism, Lausanne, Switzerland, *Critical lines and short-range correlations in a frustrated spin-1 chain*
 - 07/16 Swiss Physical Society Meeting, Lugano, Switzerland, *Dimerization transitions in spin-1 chains*
 - 07/16 Swiss Workshop on Materials with Novel Electronic Properties 2016, Les Diablerets, Switzerland, *Dimerization transitions in spin-1 chains*
 - 05/16 Japan-Swiss Workshop 'Trends in Theory of Correlated Materials', PSI, Villigen, Switzerland, *Dimerization transitions in spin-1 chains*
 - 10/14 Japan-Swiss Workshop 'Trends in Theory of Correlated Materials', Tokyo, Japan, *Berry phase investigation of spin-S ladders*
 - 07/14 Swiss Physical Society Annual Meeting, Fribourg, Switzerland, *Berry phase investigation of spin-S ladders*

Poster presentations (8):

- 09/21 Quantum Field Theory at the Boundary, Mainz, Germany, *Boundary critical phenomena in the 4-state Potts model*
- 04/21 Korrelationstage 2021, Dresden, Germany, *Chiral transitions in chains of Rydberg atoms*
- 10/18 Topological phases in condensed matter and cold atom systems, Cargèse, Corsica, France, *A comb tensor network*
- 09/17 Korrelationstage 2017, Dresden, Germany, *Dimerization and exotic criticality in spin-S chains*
- 09/16 8th International Conference on Highly Frustrated Magnetism, Taipei, Taiwan, *Dimerization transitions in spin-1 chains*
- 01/15 Theory Winter School on New Trends in Frustrated Magnetism, Tallahassee, Florida, USA, *Frustration and spontaneous dimerization in spin-1 chain*
- 10/13 School on Advanced Algorithms for Correlated Quantum Matter, Würzburg, Germany, *Berry phase investigation of spin-S ladders*
- 07/13 Swiss Workshop on Materials with Novel Electronic Properties, Les Diablerets, Switzerland, *Berry phase investigation of frustrated quantum magnets*

Event organization:

05/25-07/26 **Lorentz workshop** "Automated Reasoning for Quantum Mechanics" together with Alfons Laarman, Kuldeep Meel, and Robert Wille, Leiden, The Netherlands

07/24-01/25 Chair of the focus session "Tensor Networks: bridging quantum and classical computing" at **NWOPhysics-2025** (former Physics@Veldhoven)

05/22-08/25 The founder and the main organizer of **Delft Many-Body Workshop Series** (11 workshops); <https://nchepiga.github.io/homepage/workshop>; acquired funding from EPSRC (UK) and NWO (NL)

07/23-07/24 Topic chair of the International conference of Magnetism ICM2024, Bologna, Italy

10/23 The organizer of the IQTN/EPSRC funded workshop "Tensor networks for constrained systems" (<https://iqtn.phys.strath.ac.uk>)

02/22-05/24 Member of the program committee of International Quantum Tensor Network meetings (<https://iqtn.phys.strath.ac.uk>)

Refereeing for journals:

Nature, Nature Reviews, Nature Communications, Communications Physics;
Physical Review Letters, Physical Review Research, Physical Review B;
SciPost Physics; New Journal of Physics

Referee and panel member for funding agencies:

- Swiss National Science Foundation (SNSF)
- US Department of Energy (DOE)
- Dutch Research Council (NWO)
- Belgian Fund for Scientific Research (F.R.S.-FNRS)
- Austrian Science Fund (FWF)

Teaching & supervision:

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|---------------|---|
| 12/24-06/25 | Lecturer for the course Quantum Mechanics 2 (BSc) at TUDelft |
| 12/23-12/25 | Guest lectures in Advanced Statistical Mechanics (MSc) at TUDelft |
| 05/23 | Lecturer for the course on ' Quantum Phase Transitions ' at DRSTP postgraduate school, Callantsoog, Netherlands |
| 02/22 – 06/25 | Lecturer in ' Mechanics and Relativity ' (BSc) at TUDelft |
| 02/22 – 06/25 | Lecturer in ' Fairy Tails of Theoretical Physics ' (MSc) at TUDelft
(advanced theory lectures on: supersymmetry; frustrated magnetism; duality) |
| 09/14 – 01/17 | Teaching Assistant in ' Physique Statistique I ' at EPFL |
| 02/16 – 06/16 | Teaching Assistant in ' Mathematical Methods for Physicists ' at EPFL |
| 02/15 – 06/15 | Teaching Assistant in ' Physique Statistique II ' at EPFL |
| 09/13 – 12/13 | Teaching Assistant in ' Statistiques et probabilités ' at EPFL |
| 11/07 – 06/11 | High School Teacher in Advanced Physics and Mathematics at Private Boiko School, Ukraine |
| 08/08 – 08/09 | Summer School Teacher for Granted Youth (supported by Kharkov City Council) |

Dutch courses on teaching skills (UTQ):

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|-------|--|
| 04/21 | DEVELOP, TU Delft, ~40 hours on the development of own courses from scratch |
| 05/21 | SUPERVISE, TU Delft, ~40 hours on how to manage the group and supervise students |

Supervision

PhD students:

- Warre Missiaen, PhD thesis on quantum critical properties of interacting Majorana fermions, funded by NWO summit grant, TU Delft, Netherlands
- Julien Fitouchi, PhD thesis on unusual Mott transitions, funded by Julian Schwinger foundation, TU Delft, Netherlands
- Pietro Richelli, PhD thesis on constrained tree tensor networks, TU Delft, Netherlands

starting in October 2026 in Oxford:

- William Worthy
- Maximillian Murphy
- Adonis Haxhijaj

postdoc

- Jose Soto Garcia

Undergraduate students:

Alumni:

graduate:

- Bowy La Rivière, thesis “QUANTUM CRITICALITY IN FRUSTRATED HALDANE CHAINS AND (DOPED) QUANTUM LOOP MODELS” TU Delft, Netherlands
- Jose Soto Garcia, thesis “QUANTUM CRITICAL PROPERTIES OF RYDBERG ARRAYS IN AND OUT OF EQUILIBRIUM”, TU Delft, Netherlands

Undergraduate:

- Niels Pronk, “Numerical investigation of frustrated Haldane chain with single-ion anisotropy” TU Delft
- Filip Sfetcu „A Bosonization Study on Frustrated Zig-Zag Ladders“, TU Delft
- Ioannis Angelo Tassioulas „Study of Critical Behaviour in the Interacting Majorana-Chain Model Using Numerical Methods“, TU Delft
- Rik Mulder, “Exact zero modes in chains of interacting Majorana fermions”, TU Delft
- Luka van der Heiden, “Anisotropic spin chains with $S>1$ ”, TU Delft
- Jesse van der Kooi, „Dynamics across deconfined quantum criticality in 1D” TU Delft
- Daan van der Veer, „Kibble-Zurek drive into a floating phase” TU Delft
- Wesley Brouwer “The effect of next-to-leading-order interactions in Majorana chains”; supervised together with Dr. Jonas Thies (Math department); TU Delft
- Bernhard Luescher, intern, “Critical properties of the chiral Ashkin-Teller model”, TU Delft

co-supervised:

- Ivo Maceira’s PhD project on chiral transitions in Rydberg atoms, EPFL, Switzerland (co-supervised);
- Randy Sawaya’s PhD project on Hubbard model with long-range interactions, University of California Irvine (co-supervised);
- Robin Kaeche’s MSc thesis on critical Ising chains, EPFL (co-supervised);
- Guillaume Meyrat’s MSc project on quantum dimer model, EPFL (co-supervised);
- Samuel Gozel’s MSc thesis on dynamics in spin-3/2 chain, EPFL (co-supervised)

Member of committees:

- 09/23 Member of the poster prize committee at CT.QMAT 2023 school, Leipzig, Germany
- 08/23 Member of the poster prize committee at NG SCES 2023, Lido di Fermo, Italy
- 01/23 Member of the Minerva prize 2022 committee, The Netherlands
- 06/22 Member of the poster prize committee at Highly Frustrated Magnetism 2022, Paris, France
- 01/20 Member of the poster prize committee at Physics@Veldhoven 2020, Veldhoven, Netherlands
- 04/24 Appointment committee for a Delft Technology Fellowship at QuTech, TUDelft
- 07/21 Appointment committee for an assistant professor position at Kavli Institute, TUDelft
- 10/23-09/25 Member of the work-group “Education and outreach”, TUDelft

PhD committees:

- 09/26 Tianyue Huang, EPFL (Switzerland, supervisors A.Lauchli and F.Mila)
- 11/25 Coraline Letouze, Sorbonne University, France (supervisors G. Radtke and B. Lenz)
- 09/25 Vladimir A. Zakharov, University of Leiden, Netherlands (supervisor C.Beenakker)
- 06/25 Yann in 't Veld, Radboud University, The Netherlands (supervisor M. Roesner)
- 12/24 David Aceituno Chávez, KTH Stockholm, Sweden (supervisor J.Bardarson)
- 06/24 Alvaro Donis Vela, University of Leiden, Netherlands (supervisor C.Beenakker)
- 03/21 Member of Schelto Crone’s, University of Amsterdam, Netherlands (supervisor – P.Corboz)

Undergraduate defense committee:

- 12/24 Saqar Khaleefah, TU Delft, Netherlands (supervisor – E.Greplova)
- 07/24 Matthijs Ates, TU Delft, Netherlands (supervisor – J.Thijssen)
- 06/24 Esther Teng, TU Delft, Netherlands (supervisor – S.Otte)
- 06/24 Sander de Bruyn, TU Delft, Netherlands (supervisors – N.Budko, J.Thijssen)
- 12/23 Member of Pelle Poelmann's defense committee at TU Delft, Netherlands (supervisor – A.Artaud)
- 11/22 Expert at Luka van der Heiden's defence, TU Delft, Netherlands (supervisor – E.Pulvirenti)
- 07/22 Expert at Huang Tianyue's MSc defence, EPFL, Switzerland (supervisor – A.Lauchli)
- 07/22 Expert at Baptiste Demazure's MSc defence, EPFL, Switzerland (supervisor – F.Mila)
- 07/21 Expert at Bernhard Luescher's MSc defence, EPFL, Switzerland (supervisor – F.Mila)
- 06/21 Member of Isabel Postmes' MSc defense committee at TU Delft, Netherlands (supervisor – S.Conesa-Boj)

Outreach, volunteer and mentoring activities:

- 02/26 **Press release** on the publication “*Ising versus infinite randomness criticality in arrays of Rydberg atoms trapped with non-perfect tweezers*”
- 11/25 **Keynote** speaker at Øredev Developers conference, Malmo, Sweden
- 09/25 **Press release** on the ERC STG award by the University of Oxford
- 02/24 **Press release** on publication of “*Tunable quantum criticality in multi-component Rydberg arrays*”
- 12/23 **SURF** advanced computing day, public talk “Challenging quantum simulators with classical computers”
- 11/23 Article by Kenna Hughes-Castleberry for “**Women of Quantum Technology**”, <https://www.insidequantumtechnology.com/news-archive/women-of-quantum-technology-dr-natalia-chepiga-of-delft-university-of-technology/>
- 10/23 DelftBlue **HPC summit**, public talk “*9 ½ phases of interacting Majorana chains*”
www.tudelft.nl/en/events/2023/dcse/user-summit-2023
- 05/2023 **Interview** in honor of DelfBlue HPC anniversary
www.tudelft.nl/en/stories/articles/supercomputing-power-for-racing-cars-and-quantum-states
- 01/2022 **Interview** for Nederlands Tijdschrift voor Natuurkunde

Since 2023: The member of Grete Hermann network of females in condensed matter physics

Since 2022: **The mentor** in EPFL's Alumni mentoring program

2016-2020 4 public lectures at the Boiko School, Kharkiv, Ukraine:

- 08/10 Environmental volunteer program in Vichy, France
- 07/10 Camp leader in the international volunteer project, Lyubotin, Kharkiv district, Ukraine
- 08/09 Volunteer in Summer Camp in Spangenberg, Germany
- 11/07 Team leader in Ukraine-China exchange program in Shitzyatjuan, China

References:

1. Prof. Frédéric Mila, École Polytechnique Fédérale de Lausanne, frederic.mila@epfl.ch
2. Prof. Nicolas Laflorencie, LPT Toulouse, laflor@irsamc.ups-tlse.fr
3. Prof. Frank Verstraete, Cambridge and UGhent, frank.verstraete@ugent.be
4. Prof. Philippe Corboz, University of Amsterdam, P.R.Corboz@uva.nl
5. Prof. Paul Fendley, University of Oxford, paul.fendley@physics.ox.ac.uk
6. Prof. Fabian Essler, University of Oxford, fabian.essler@physics.ox.ac.uk